6A. ASHRAE-referenced calculation sheet

Calculated substitutions are shown below. Engineering review required; not ASHRAE certification.

1. OA specific volume

v = 287.042*(T+273.15)*(1+1.607858*w)/P

287.042*(35.000+273.15)*(1+1.607858*0.0069851)/101325.0

Result: 0.88275749 m3/kg_da

Reference: Ideal-gas moist-air approximation; see ASHRAE Handbook Fundamentals, Psychrometrics

2. Supply dry-air mass flow

m_s = Vdot_s / v_oa

4410.00/1000 / 0.8827575

Result: 4.99570950 kg_da/s

Reference: Psychrometric mass balance

3. Exhaust dry-air mass flow

m_e = Vdot_e / v_ra

4270.00/1000 / 0.8543742

Result: 4.99781020 kg_da/s

Reference: Psychrometric mass balance

4. Minimum heat-capacity rate

C_min = min(m_s*cp_s, m_e*cp_e)

min(4.995710*1.018992, 4.997810*1.023291)

Result: 5.09058958 kW/K

Reference: ASHRAE Ch.26 Eq. (2a) capacity-rate approach, p.26.2

5. Sensible recovery

q_s = eps_s*C_min*(T_oa-T_ra)

0.50000*5.090590*(35.000-24.000)

Result: 27.99824271 kW

Reference: ASHRAE Ch.26 Eqs. (2a), (5a), pp.26.2-26.3

6. Supply recovery outlet temperature

T_2 = T_1 - q_s/C_s

35.0000-27.998243/5.090590

Result: 29.50000000 deg C

Reference: ASHRAE Ch.26 Eq. (4a), p.26.3; constant cp approximation

7. Exhaust recovery outlet temperature

T_4 = T_3 + q_s/C_e

24.0000+27.998243/5.114216

Result: 29.47459083 deg C

Reference: Energy conservation; ASHRAE Ch.26 Eq. (4b), p.26.3

8. Water vapor transfer

m_w = eps_L*min(m_s,m_e)*(w1-w3)

0.50000*4.995710*(0.0069851-0.0092965)

Result: -0.00577346 kg_w/s

Reference: ASHRAE Ch.26 Eqs. (2b), (5c), pp.26.2-26.3

9. Supply humidity ratio after recovery

w2 = w1 - m_w/m_s

0.0069851-(-0.005773462/4.995710)

Result: 0.00814080 kg_w/kg_da

Reference: ASHRAE Ch.26 Eq. (4c), p.26.3

10. Exhaust humidity ratio after recovery

w4 = w3 + m_w/m_e

0.0092965+(-0.005773462/4.997810)

Result: 0.00814128 kg_w/kg_da

Reference: Moisture conservation; ASHRAE Ch.26 Eq. (4d), p.26.3

11. Total recovered energy

q_total = m_s*(h1-h2)

4.995710*(53.134503-50.483824)

Result: 13.24202326 kW

Reference: ASHRAE Ch.26 Eq. (5d), p.26.3

12. Coil target duty (signed)

q_coil = m_s*(h2-h_supply)

4.995710*(50.483824-38.518052)

Result: 59.77752319 kW

Reference: Steady-flow enthalpy balance; not a coil selection

Balance checks (residual and absolute tolerance)

PASS: Sensible energy closure — 3.5527137e-15 kW (limit +/-1e-07)

PASS: Moisture conservation — -1.7347235e-18 kg/s (limit +/-1e-08)

PASS: Approximate total enthalpy closure — -0.00027285996 kW (limit +/-0.3310574)